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Ask Gemini

Alex Maddy S
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CO5-AT2-Short Answers Test

QUESTIONS

Q1

Define a graph.
2 marks

Q2

What are the major types of graphs?
2 marks

Q3

Differentiate between directed and undirected graphs.
2 marks

Q4

What is a weighted graph?
2 marks

Q5

What is a topological sort?
2 marks

Q6

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$.
2 marks

Q5

What is a topological sort?

2 marks

What is a topological sort?

Your Answer

Topological sort is a linear ordering of vertices of vertices in a directed acyclic graph such that for every directed edge, one vertex comes before another vertex in the ordering.

Save Answer

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Q6

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks

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marks

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.

Your Answer

A valid topological ordering is A,B,C,E,D,F. This ordering is valid because every prerequisite node comes before its dependent node according to the edges A-C,B-C,C-D,B-E,E-F.

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